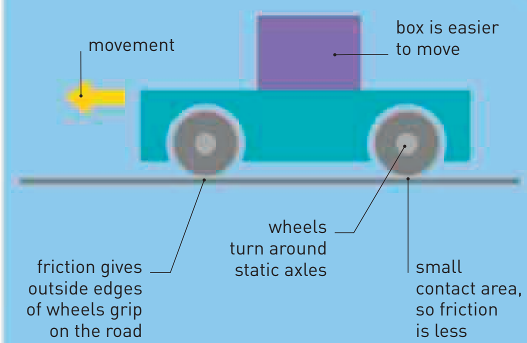

# THE STORY OF THE WHEEL

**THIS SIMPLE INNOVATION HAS MOVED ARMIES, CARRIED LOADS, AND POWERED INDUSTRIES**

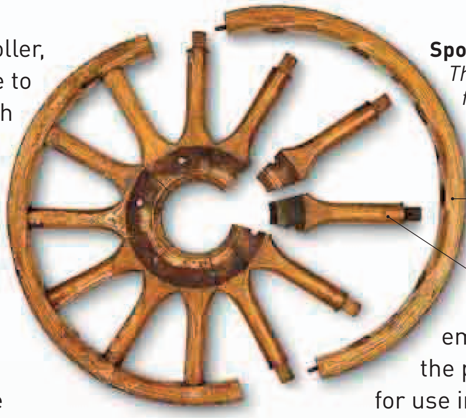
**One of the most important inventions in history, the wheel allowed the transportation of loads over long distances, revolutionized early warfare, and made the development of the first mechanized processes possible. It opened up the globe to human exploration and revolutionized industry.**




### WHEELS AND FRICTION

The force needed to pull a load pressing down directly on the ground is increased by the friction or "rolling resistance" between the load and the ground. The use of wheels resolves this problem. Since only a small part of the wheel is in contact with the ground at any one point in time, the rest of it can rotate freely, without being impeded by friction. The little friction that remains allows the wheel to grip the ground without sliding. Wheels are mounted on sturdy shafts, called axles, which facilitate the rolling motion.

The earliest wheel, the logroller, was used by neolithic people to transport heavy weights, such as large stones used in the construction of megaliths. By 3500 BCE, the logroller was adapted to create the first true wheels—solid disks of wood connected by an axle. These wheels, however, were very heavy. Lighter, spoked wheels were invented around 1600 BCE. The more hard-wearing, iron-rimmed wheels came around 800 years later, making for faster, more durable vehicles suitable for battle and long-distance transportation. Wheels steadily evolved, using materials such as iron and steel as they were developed. Modern wheels use high-tech alloys of titanium or aluminum that are light and allow vehicles to move faster, using much less power.




**Spoked wheel construction**
*The spokes of a wheel distribute the force applied to a vehicle evenly around its rim. As the wheel rotates, each spoke shortens slightly.*

### THE WHEEL IN INDUSTRY

Beginning with the potter's wheel around 4500 BCE, the wheel was also adapted for use in industrial processes. By 300 BCE, watermills were employed in Greece to harness the power of water, via a turbine, for use in milling. By the time of the Industrial Revolution, the wheel appeared in one form or another in almost all industrial machinery. Gears (toothed wheels) and cogs were used in the Antikythera mechanism—an astronomical calculating machine created in Greece around 100 BCE—but it is possible they were used earlier in China. Gears and cogs eventually became common components of machines as diverse as clocks and automobiles. Yet there were some cultures where the wheel did not feature as prominently. Some ancient civilizations of Central America and Peru did not develop wheels, or, as in the case of the Aztecs of Mexico, used them only in children's toys.

***c*.100 BCE**
**Wheelbarrow**
The Chinese create a wheelbarrow with a large central wheel, which makes all the weight fall on the axle. Easy to push, each wheelbarrow can carry up to six men.


"Wooden ox" wheelbarrow


Chinese spinner

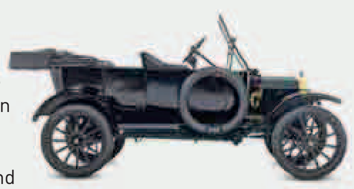
***c*.1035**
**Spinning wheel**
In China, a hand-crank-operated driving wheel is added to a hand-spindle, automating it and allowing multiple spindles to be operated simultaneously.

**1845**
**Vulcanized rubber tire**
Robert Thomson uses vulcanized rubber—invented by Charles Goodyear—to make pneumatic (air-filled) tires, which are lighter and harder to wear out.

**1848**
**Mansell wheel**
The quieter and more resilient Mansell railroad wheel has a steel central boss (hub), surrounded by a solid disk of 16 teak segments.


Gazelle steam engine

**1910**
**Early automobile spoked wheel**
Earliest automobile wheels have wooden spokes, which are more suitable for narrow tires, but tend to warp and crack.


Ford Model T

**1915**
**Radial tire**
Patented by Arthur Savage, radial ply tires are made of rubber-coated steel or polyester cords. They are now the standard tire for almost all cars.


1960s Mini

**2010**
**Modern wheel types**
Ultra-lightweight racing bicycles use composite carbon spokes, while car wheels are made of magnesium, titanium, or aluminum alloys.


High-tech racing bike